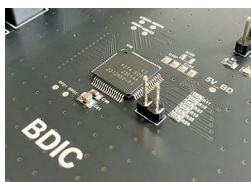


unparalleled total cost of ownership (TCO) superiority. The memory solution caters to the rising needs of the generative AI landscape, accommodating multimodal and multitrillion-parameter AI models.

Micron Technology, Inc.
<https://www.micron.com>

World's first 14-channel battery diagnosis IC for EVs and ESSs

Autosilicon has launched the world's first 14-channel battery diagnosis IC (BDIC) for electric vehicles (EV) and energy storage systems (ESS). BDIC significantly enhances measurement precision, operational current, and compactness



compared to electrochemical impedance spectroscopy (EIS) equipment. A single BDIC assesses up to 14 cells' impedance, which helps diagnose, predict, and inspect battery cells' health, including state of health (SoH) and remaining useful lifetime (RUL). An algorithm and software measure the internal temperature of individual cells using BDIC, eliminating the need for external temperature sensors. This enables the early detection of internal or localised hot-spots in battery cells, mitigating the risk of fires or explosions. BDIC enhances the battery diagnosis function of existing battery management systems (BMS) and lays the foundation for battery cell grading necessary for the reutilisation of discarded batteries. It can be effectively employed in various battery diagnosis fields and battery-as-a-service (BaaS) businesses.

Autosilicon
<https://www.autosilicon.com>

Ultra-sensitive camera for precision monitoring and remote operation

Canon's MS-500 is the world's first ultra-high-sensitivity interchangeable-

lens camera (ILC) equipped with a 25.4mm single photon avalanche diode (SPAD) sensor featuring the world-leading pixel count of 3.2 megapixels. The camera offers low-light capability,



coupled with broadcast lenses optimised for telephoto focal lengths, making

it a prospective choice for precision monitoring applications. The bayonet lens mount empowers the camera to recognise and capture subjects that are kilometres away. The B4 mount, integrated 1.3x optical magnifier, and 25.4mm type SPAD sensor unite to ensure precise colour monitoring of distant subjects under both well-lit conditions and darkness. The camera is robust and reliable, functioning 24/7 across temperatures from -20°C to 45°C. It effortlessly toggles between manual and automatic modes for the detachable IR cut filter, capturing extra IR light for post-dark shooting. Additionally, the convenience of remote operation is also available.

Canon Inc.
<https://www.canon-europe.com>

Enhancing IoT with capacitor-based backup power innovation

Blues have introduced Scoop, a capacitor-based backup power add-on board. Unlike lithium polymer batteries (LiPOs), this capacitor functions across



wider temperature ranges, boasts a lower self-discharge rate, and sidesteps

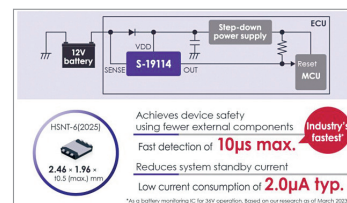
the shipping restrictions associated with lithium batteries. Scoop's stand-out advantage lies in its power outage detection capability, which is vital for mission-critical scenarios susceptible

to substantial consequences from power interruptions. Scoop facilitates the IoT device in making a seamless transition to backup power, detecting the alteration in charge, and promptly sending an alert for power loss. The board has the capacity to fuel up to 200 connections to Nethub on a complete charge, contingent on the wireless coverage available and data transmission volume in the vicinity. This encompasses nearly four hours of uninterrupted operation.

Blues
<https://shop.blues.io>

Fast and efficient automotive battery monitoring ICs

ABLIC has introduced the S-19114 Series, a line of automotive high-withstand voltage battery monitoring ICs known for their fastest voltage detection response, particularly suitable for challenging conditions. These ICs have a compact design with an ultra-compact automotive package, reducing its footprint by approximately 20% compared to previous models. The voltage detection response



time has been significantly improved, enhancing its suitability for demanding environments. They operate with low current consumption, ensuring continuous battery monitoring. The wide-rated voltage range of the monitoring voltage input pin (-30V to 45V) eliminates the need for conventional dividing resistors that typically lead to higher standby currents. This advancement, combined with reduced capacitors and smaller package sizes, results in substantial space savings in automotive applications. The Series finds utility in automotive battery voltage detection, battery management systems (BMS), onboard chargers, and DC/DC converters.

ABLIC
<https://www.ablic.com>